1 EXECUTIVE SUMMARY & PRODUCT VISION

Nearzy is a modern, high-performance hyperlocal marketplace built specifically for Tamil Nadu. It seamlessly connects residents with verified local service professionals (electricians, plumbers, AC technicians, beauticians, carpenters) within a precise **30-kilometer geographical radius**. Featuring real-time Leaflet maps, an intelligent bilingual AI assistant, and instant role-based dashboards, Nearzy digitizes neighborhood trades with zero friction.

2 GITHUB, HOSTING & INFRASTRUCTURE

COMPONENT	CONFIGURATION / PROVIDER	KEY IMPLEMENTATION DETAILS
GitHub Repository	`JoTech17/Nearzy` (Git)	Version controlled on GitHub with dual synced branches: `master` (primary production) and `main`. Clean atomic commit history.
Frontend Hosting	Vercel / Netlify / Vite SPA Server	Static Client-Side SPA build served via global Edge CDN. Configured with `vercel.json` rewrite rules (`/*` → `/index.html`) for dynamic routing.
Backend & Database	Supabase Cloud (BaaS)	Managed PostgreSQL cluster (`ewczncwreeangxalnwqz.supabase.co`) with automated connection pooling, SSL encryption, and JWT authentication.
AI Inference Host	Google AI Gemini API	RESTful integration with `gemini-2.5-flash` providing ultra-low latency, natural language semantic search, and spatial filtering.
Security & Protocols	HTTPS / TLS 1.3 / JWT / Row Level Security	Row Level Security (RLS) policies on every Postgres table; encrypted bearer token authorization on all privileged database transactions.

3 HIGH-LEVEL SYSTEM ARCHITECTURE



4 CORE TECHNOLOGY STACK MATRIX

Frontend Layer — UI / UX

- **React 19.2.8** (Modern functional components)
- **Vite 8.3.4** (Sub-second build & HMR)
- **Tailwind CSS 3.4** (Design system)
- **Framer Motion 14.4** (Route transitions & physics)
- **Lucide React** (Pixel-perfect icon library)

Backend & Storage — Database

- **PostgreSQL 15+** (Relational core)
- **Supabase JS SDK 2.118** (Client queries)
- **Postgres Triggers** (Auto-profile creation)
- **Row Level Security (RLS)** (Role isolation)
- **pg_trgm & UUID** (Fuzzy search & unique IDs)

Spatial & Intelligence — AI & Maps

- **Google Gemini Flash** (Tamil & EN Assistant)
- **Haversine Formula** (30 km radius math)
- **Leaflet 1.9 & React-Leaflet** (Map view)
- **OpenStreetMap Tiles** (Statewide raster tiles)
- **38 Districts Dataset** (760+ localized shops)

5 DATABASE ARCHITECTURE & AUTOMATED TRIGGERS

The relational database runs on Supabase PostgreSQL. It strictly enforces foreign keys, cascades, and security policies. User authentication directly activates database-level triggers to guarantee zero orphaned records.

TABLE NAME	PRIMARY COLUMNS & RELATIONS	PURPOSE & RLS POLICIES
`auth.users`	`id (UUID PK), email, encrypted_password, raw_user_meta_data`	Supabase internal authentication store. Manages encrypted user credentials, email confirmation tokens, and sessions.
`public.profiles`	`id (UUID FK → auth.users), display_name, email, phone, preferred_language`	Public profile entity. Populated instantly via `handle_new_user()` trigger. Viewable by public, updatable only by the owner.
`public.user_roles`	`id (PK), user_id (FK → profiles), role ('customer' | 'business_owner' | 'admin')`	Role-based access control (RBAC). Determines permissions for managing listings, claiming shops, or moderating content.
`public.businesses`	`id (PK), category_id, name, slug, address, locality, city, latitude, longitude, phone, is_verified, created_by`	Stores 760+ localized service businesses across Tamil Nadu. Indexed with geographic coordinates for fast spatial querying.
`public.services` & `bookings`	`id, business_id, name, price, duration_minutes | customer_id, booking_date, status`	Transactional booking subsystem. Allows customers to book appointments and track status (pending, confirmed, completed).

Automated Signup Database Trigger: public.handle_new_user()

When a user signs up through `supabase.auth.signUp()`, the trigger `on_auth_user_created` immediately executes `public.handle_new_user()` on PostgreSQL. It extracts the display name, email, and preferred language from metadata and inserts atomic records into `public.profiles` and `public.user_roles`. This ensures frontend code never has to perform manual profile bootstrapping.

6 AI ASSISTANT & 30 KM SPATIAL GEOMETRY ENGINE

Google Gemini AI Engine

`Gemini Flash`

- **Context-Aware Natural Language**: Trained with platform system instructions containing catalog rules, contact details, and Tamil Nadu geography.
- **Full Tamil & English Fluency**: Understands colloquial Tamil queries (e.g. *"காரைக்குடியில் உள்ள எலக்ட்ரீஷியன்கள் காட்டு"*) as accurately as English.
- **Intent Recognition**: Automatically detects if the user is seeking a service, checking prices, asking how to list a business, or seeking customer care.

30 KM Haversine Spatial Radius

`Mathematical Core`

- **Spherical Trigonometry**: Uses the great-circle distance formula to compute exact curvature distance across Earth's surface:

$$d = 2R \cdot \arcsin\left(\sqrt{\sin^2(\Delta\phi/2) + \cos(\phi_1)\cdot\cos(\phi_2)\cdot\sin^2(\Delta\lambda/2)}\right)$$

- **Strict Boundary Enforcement**: When a user searches for a specific locality (e.g. Karaikudi), the AI strictly filters listings within ≤ 30 km radius.

7 BILINGUAL ENGINE & MICRO-ANIMATIONS

Zero-Dependency Bilingual System (EN / தமிழ்)

Nearzy features a high-speed, zero-dependency translation context (`LanguageContext.jsx`). All UI labels, service categories, buttons, legal policies, and AI recommendations switch seamlessly between English and Tamil with 0ms delay.

Framer Motion Route Transmissions

Integrated with `framer-motion 14.4` and `AnimatePresence mode="wait"`. Every route navigation glides smoothly using custom cubic-bezier curves (`[0.16, 1, 0.3, 1]`), accompanied by spring-physics dialogs and staggered card entrances.

8 KEY QUESTIONS & ANSWERS (EVERYTHING YOU NEED TO KNOW)

Q1 What is Nearzy and what core problem does it solve?

Nearzy is an on-demand hyperlocal service discovery platform for Tamil Nadu. It solves the fragmentation of finding verified neighborhood electricians, plumbers, carpenters, and beauty technicians by providing real-time 30 km radius matching, bilingual navigation, and direct booking.

Q2 Which tools and agent were used to create this website?

The website was developed using **Google DeepMind Antigravity** (Advanced Agentic AI Coding Assistant) paired with the human engineer. Development tools included **VS Code**, **Node.js v20+**, **Vite**, **Git CLI**, and **Supabase CLI** for schema migrations.

Q3 Where is the source code hosted and how is version control managed?

The complete source code is hosted on GitHub at `https://github.com/JoTech17/Nearzy.git`. Version control uses dual synchronized branches (`master` and `main`) with atomic commits and semantic feature tags for instant redeployment.

Q4 Where is the website deployed and how does routing work?

The frontend is deployed as a high-speed Single Page Application (SPA) on modern cloud hosting (Vercel / Netlify Edge). Routing is powered by React Router v7 with rewrite rules configured in `vercel.json` to ensure all deep links (e.g., `/register-business`, `/privacy`, `/admin`) load smoothly without 404 errors.

Q5 What database is used and how is user information stored?

Nearzy uses **Supabase PostgreSQL**. User authentication is managed in `auth.users`, which automatically synchronizes with `public.profiles` via the PostgreSQL trigger `handle_new_user()`. Role separation (Customer, Business Owner, Admin) is maintained in `public.user_roles` with Row Level Security.

Q6 How does the AI Assistant provide accurate answers and 30 km recommendations?

The assistant is powered by **Google Gemini Flash** via REST API. When a query is received, it resolves the location coordinates against a master dataset of 38 Tamil Nadu districts and applies the Haversine formula to return exclusively verified providers within 30 km.

Q7 How can an administrator verify registered profiles in the database?

Administrators have three convenient methods: (1) Running `npm run check:profiles` in the terminal to inspect live database records, (2) Visiting the **Registered Profiles** tab on the Nearzy Admin Console (`/admin`), or (3) Checking the `profiles` table directly in the Supabase Dashboard.

Q8 How does the bilingual English and Tamil feature work?

The bilingual engine is powered by React's Context API (`LanguageContext.jsx`). All navigation, category titles, service details, and legal pages are stored as clean bilingual dictionaries with instant one-click switching and zero page reload delay.

Q9 How was the UI made modern, smooth, and interactive?

We integrated **Tailwind CSS** for a sleek red-black-white visual hierarchy, coupled with **Framer Motion** for spring physics modals, smooth page route transmissions, interactive Leaflet map markers, and animated drawer menus.

Q10 What are the next planned enhancements for Nearzy?

Planned milestones include Razorpay UPI payment gateway settlement, SMS / WhatsApp OTP notifications via Twilio / Gupshup, and a progressive web app (PWA) offline mobile service provider portal.